

1. Introduction

This report outlines the initial activities and current progress of our project for the Bosch Future Mobility Challenge. The primary focus during this phase was on setting up the vehicle, understanding the provided software framework, and achieving stable manual control using the demo code. We have initiated autonomous functionality through stop sign detection, which is capable of stopping the vehicle, though accuracy improvements are ongoing. In parallel, the simulator environment was successfully set up to visualize the track and vehicle connection. This report summarizes the work completed so far and defines the upcoming tasks toward full autonomy.

2. Planned Activities

The planned activities are as follows:

- Architecture
- Project Plan
- Manual Vehicle Control
- Simulator Setup
- Stop Sign Detection

Architecture

Status: Completed

Implementation:

The system architecture was studied and implemented using the Bosch-provided framework. Communication between the Raspberry Pi, STM32 Nucleo, sensors, actuators, and simulator was verified.

Difficulties:

Understanding the interaction between multiple software and hardware components required careful analysis of the framework.

Project Plan

Status: Completed

Implementation:

Project development was aligned with the defined architecture and competition timeline, focusing on step-by-step progress from manual control to autonomous features.

Difficulties:

Adapting the plan to initial setup challenges required minor timeline adjustments.

Manual Vehicle Control

Status: Completed

Implementation:

The vehicle was successfully operated in manual mode using the provided demo code. Steering and speed control through the dashboard were verified.

Difficulties:

Initial calibration and configuration required repeated testing.

Simulator Setup

Status: Completed

Implementation:

The simulator environment was successfully configured. The track map is visible, and the vehicle is connected correctly. Autonomous task implementation in the simulator has not yet started.

Difficulties:

Simulator configuration and synchronization with the framework required careful setup.

Stop Sign Detection

Status: Ongoing

Implementation:

A stop sign detection module was implemented on the real vehicle. When switched from manual to autonomous mode, the vehicle is able to detect a stop sign and stop. Detection accuracy is currently limited under certain lighting conditions.

Difficulties:

Improving robustness and consistency in real-time operation remains challenging.

3. General Status of the Project

The project has progressed steadily with successful vehicle setup, architecture implementation, simulator readiness, and stable manual control. Autonomous functionality has been initiated through stop sign detection, while further improvements are ongoing. Traffic light detection and lane detection are planned for upcoming phases to achieve full autonomous operation.

4. Upcoming Activities

1. Improve accuracy and reliability of stop sign detection.
2. Continue development of traffic light detection.
3. Start lane detection and tracking for straight and curved paths.

